



UNIVERSITY OF  
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# Superconductivity

## Lecture 10 – “Strong Fields”

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# Linearised Ginzburg-Landau equation

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# Linearisation

How to solve?

$$\frac{1}{2m^*} (-i\hbar\nabla + e^*\mathbf{A})^2 \psi + a\psi + b\psi|\psi|^2 = 0 \quad (1)$$

Or, where to apply?

- Linearise the equation
- where  $\psi$  is small ( $H \lesssim H_{c2}$ )

So, let's

- Drop the term  $\psi|\psi|^2$  off
- Apply  $\mathbf{B}_a = \mu_0 \mathbf{H}_a = (0, 0, B_a)$

You may choose the Landau gauge  $\mathbf{A} = (0, xB, 0)$ .

## Linearised GL equation

By inserting  $\mathbf{A} = (0, xB, 0)$  into

$$\frac{1}{2m^*} (-i\hbar\nabla + e^*\mathbf{A})^2 \psi + a\psi = 0, \quad (2)$$

One obtains

$$\left( -\frac{\hbar^2}{2m^*} \nabla^2 - \frac{\hbar e^* B_a}{m^*} i x \frac{\partial}{\partial y} + \frac{e^{*2} B_a^2}{2m^*} x^2 \right) \psi = |a| \psi, \quad (3)$$

where one might need to use  $\nabla = \hat{e}_x \frac{\partial}{\partial x} + \hat{e}_y \frac{\partial}{\partial y} + \hat{e}_z \frac{\partial}{\partial z}$ .

Looks a bit complicated? One can simplify the above by noting

- GL eq. was told to be Schrodinger eq.-like except the nonlinear term (which we dropped!),
- The linearised eq. contains  $-i\hbar\partial/\partial y$  term and
- $eB_a/m^*$ , that has a frequency dimension, is here and there.

## Simplify linearised GL equation

By defining a cyclon frequency  $\omega_c \equiv eB_a/m^*$ ,

$$\left( -\frac{\hbar^2}{2m^*} \nabla^2 - \frac{\hbar e^* B_a}{m^*} i x \frac{\partial}{\partial y} + \frac{e^{*2} B_a^2}{2m^*} x^2 \right) \psi = |a| \psi, \quad (4)$$

can be written as

$$\left( -\frac{\hbar^2}{2m^*} \nabla^2 - \hbar \omega_c i x \frac{\partial}{\partial y} + \frac{1}{2} m^* \omega_c^2 x^2 \right) \psi = |a| \psi \quad (5)$$

This looks the same as the Schrodinger equation for a charged particle in a magnetic field, where the solution results in Landau levels.

## Solution to the linearised equation

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## Solution (1/3)

To solve

$$\left(-\frac{\hbar^2}{2m^*}\nabla^2 - \hbar\omega_c ix \frac{\partial}{\partial y} + \frac{1}{2}m^*\omega_c^2 x^2\right) \psi = |a|\psi \quad (6)$$

Start with a trial solution,

$$\psi(\mathbf{r}) = e^{i(k_y y + k_z z)} f(x), \quad (7)$$

which gives

$$\nabla^2 \psi = e^{i(k_y y + k_z z)} \frac{d^2 f}{dx^2} - (k_y^2 + k_z^2) e^{i(k_y y + k_z z)} f \quad (8)$$

and

$$\frac{\partial}{\partial y} \psi = i k_y e^{i(k_y y + k_z z)} f. \quad (9)$$

## Solution (2/3)

One can rewrite the linearised GL equation

$$\left( -\frac{\hbar^2}{2m^*} \nabla^2 - \hbar\omega_c i x \frac{\partial}{\partial y} + \frac{1}{2} m^* \omega_c^2 x^2 \right) \psi = |a| \psi \quad (10)$$

as

$$-\frac{\hbar^2}{2m^*} \frac{d^2 f}{dx^2} + \left( \hbar\omega_c k_y x + \frac{1}{2} m^* \omega_c^2 x^2 \right) f = \left( |a| - \frac{\hbar^2 (k_y^2 + k_z^2)}{2m^*} \right) f \quad (11)$$

This can be further simplified by defining

$$x_0 \equiv -\hbar k_y / m^* \omega_c. \quad (12)$$

The resulting equation is

$$-\frac{\hbar^2}{2m^*} \frac{d^2 f}{dx^2} + \frac{m^* \omega_c^2}{2} (x - x_0)^2 f = \left( |a| - \frac{\hbar^2 (k_z^2)}{2m^*} \right) f. \quad (13)$$

The same form as the Schrodinger equation for a simple harmonic oscillator.

The solution to the harmonic oscillator

$$\left(n + \frac{1}{2}\right) \hbar \omega_c = |a| - \frac{\hbar^2 k_z^2}{2m^*} \quad (14)$$

If this were a problem of an electron in a magnetic field, then  $|a|$  would've been an energy  $E_n(k_z)$ . Then the solution becomes

$$E_n(k_z) = \left(n + \frac{1}{2}\right) \hbar \omega_c + \frac{\hbar^2 k_z^2}{2m_e} \quad (15)$$

where  $\omega_c$  should be  $eB/m_e$ .

## Implication of the solution

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# How to interpret the solution

The solution

$$\left(n + \frac{1}{2}\right) \hbar \omega_c = |a| - \frac{\hbar^2 k_z^2}{2m^*} \quad (16)$$

can be expanded as

$$\left(n + \frac{1}{2}\right) \hbar \omega_c + \frac{\hbar^2 k_z^2}{2m^*} = \frac{da}{dT} (T_c(0) - T). \quad (17)$$

For  $n = 0$  and  $k_z = 0$ , this reduces to

$$\frac{1}{2} \hbar \omega_c = \frac{da}{dT} (T_c(0) - T). \quad (18)$$

As we have applied  $H_a$ , by setting  $T \rightarrow T_c(H_a)$ , one obtains

$$T_c(H_a) = T_c(0) - \frac{1}{2 (da/dT)} \hbar \omega_c. \quad (19)$$

The critical temperature is suppressed by the applied field.

# Implication of the solution

We may rewrite the solution as

$$\frac{1}{2} \hbar \frac{e^* \mu_0 H_{c2}}{m^*} = |a| \quad (20)$$

where we have restored  $\omega_c = e^* B / m^*$ . One can use the definition  $\xi(T)$  and a flux quantum  $\Phi_0 = h / e^*$  to show

$$\mu_0 H_{c2} = \frac{\Phi_0}{2\pi \xi(T)^2}. \quad (21)$$

You may appreciate that  $H_{c2}$  is simply defined with  $\Phi_0$  and  $\xi(T)$ .

$\Rightarrow$  one flux quantum in unit area of  $2\pi \xi^2$  at  $H_{c2}$ .

One can also show that

$$\mu_0 H_{c2} = \frac{\Phi_0}{2\pi \xi(0)^2} \frac{T_c - T}{T_c}. \quad (22)$$

What does this tell us? It allows us to obtain  $\xi(0)$  by measuring temperature dependence of  $H_{c2}$ .

## Type I & II criterion

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## Critical field $H_c$

$$H_c = \frac{(da/dT)}{(\mu_0 b)^{1/2}} (T_c - T) \quad (23)$$

$$= \frac{1}{\mu_0} \frac{\Phi_0}{2\pi} \frac{1}{\xi^2} \frac{\xi}{\sqrt{2}\lambda} \quad (24)$$

$$= H_{c2} \frac{1}{\sqrt{2}} \left( \frac{\xi}{\lambda} \right) \quad (25)$$

By defining a temperature independent parameter

$$\kappa = \lambda/\xi \quad (26)$$

one can write  $H_{c2} = \sqrt{2}\kappa H_c$ . So there can be two cases where i)  $\kappa > 1/\sqrt{2}$  such that  $H_{c2} > H_c$  and ii)  $\kappa < 1/\sqrt{2}$  such that  $H_{c2} < H_c$ . The former corresponds to type-II and the latter type-I superconductors, respectively.